Math for CS & AI: Homework 8

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Problem 4.11

Let $f : S \to U$, $g : U \to S$ be holomorphic functions on two open subsets S and U of $\mathbb{C}$ such that $f(g(z)) = z$ for all $z \in U$ and $g(f(w)) = w$ for all $w \in S$. That is, f and g are inverse functions. Assume $0 \in S$ and $f(0) = 0$. Prove that there exists $r > 0$ such that for all $z \in D_r(0)$,

$$g(z) = \sum_{n \geq 1} g_n z^n,$$

where $g_n := \frac{1}{n} \operatorname{Res}(h_n, 0)$ and h_n is the following function:

$$h_n(w) = \left(\frac{1}{f(w)} \right)^n.$$

Answer:

Let $\gamma : [a, b] \to S$ denote the simple closed curve C_r. Since g is a holomorphic function, we have

$$\begin{aligned} g_n &= \frac{1}{2\pi i} \int_{C_r} \frac{g(w)}{w^{n+1}} \mathrm{d}w \\ &= \frac{1}{2\pi i} \int_a^b \frac{g(\gamma(t))}{\gamma(t)^{n+1}} \gamma'(t) \mathrm{d}t. \end{aligned}$$

For $n = 0$, we have $g_0 = 0$ because $g(0) = 0$ (as f and g are inverse functions with $f(0) = 0$).

Let $\tilde{\gamma}(t) = g(\gamma(t))$, then $\gamma(t) = f(\tilde{\gamma}(t))$. For $n \geq 1$, we perform integration by parts to rewrite the integral:

$$\begin{aligned} g_n &= \frac{1}{2\pi i} \int_a^b g(\gamma(t)) \frac{\gamma'(t)}{\gamma(t)^{n+1}} \mathrm{d}t \\ &= -\frac{1}{2n\pi i} \int_a^b \tilde{\gamma}(t) \left(\frac{1}{f(\tilde{\gamma}(t))^n} \right)' \mathrm{d}t \\ &= -\frac{1}{2n\pi i} \left[\frac{\tilde{\gamma}(t)}{f(\tilde{\gamma}(t))^n} \Big|_a^b - \int_a^b \frac{\tilde{\gamma}'(t)}{f(\tilde{\gamma}(t))^n} \mathrm{d}t \right] \\ &= \frac{1}{2n\pi i} \int_a^b \frac{\tilde{\gamma}'(t)}{f(\tilde{\gamma}(t))^n} \mathrm{d}t \\ &= \frac{1}{2n\pi i} \int_{C_r} h_n(w) \mathrm{d}w \\ &= \frac{1}{n} \operatorname{Res}(h_n, 0). \end{aligned}$$

Problem 4.12

Let t_n be the number of trees with n vertices labeled from 1 to n. Here, a tree is a connected (undirected) graph that contains no cycles. Two trees are considered the same if for every pair of vertices u and v, either there is an edge between u and v in both trees, or there is no edge between u and v in both trees. Further, let a_n be the number of rooted trees with n vertices labeled from 1 to n. That is, the number of pairs (T, r) where T is a tree and r is a vertex of T, designated as the root. Obviously, $a_n = n \cdot t_n$. Let $\widehat{A}(x)$ be the exponential generating function of a_n:

$$\widehat{A}(x) = \sum_{n \geq 1} a_n \frac{x^n}{n!}.$$

(a) Prove that

$$\widehat{A}(x) = xe^{\widehat{A}(x)}.$$

(b) Determine a closed-form formula for a_n.

(c) Determine a closed-form formula for t_n.

Answer:

For part (a), consider the root r of the tree. If we remove the root vertex r, the tree decomposes into a collection of rooted subtrees. These subtrees are unordered, which implies:

$$\begin{aligned}\widehat{A}(x) &= x \sum_{n \geq 0} \frac{\widehat{A}(x)^n}{n!} \\ &= xe^{\widehat{A}(x)}.\end{aligned}$$

For part (b), we define the holomorphic functions f and g such that $g(z) = \widehat{A}(z) = \sum_{n \geq 1} a_n \frac{z^n}{n!}$, and $f(z)$ is the inverse function of $g(z)$, which means $f(g(z)) = g(f(z)) = z$.

Since $g(w) = we^{g(w)}$ for all w, substituting $w = f(z)$ gives $z = f(z)e^z$, so $f(z) = \frac{z}{e^z}$. By the Lagrange inversion theorem we derived in Problem 4.11, we have:

$$\begin{aligned}g(z) &= \sum_{n \geq 1} \frac{z^n}{2n\pi i} \int_{C_r} \frac{e^{nz}}{z^n} \mathrm{d}z \\ &= \sum_{n \geq 1} \frac{z^n}{2n\pi i} \int_{C_r} \frac{1}{z^n} \sum_{t \geq 0} \frac{(nz)^t}{t!} \mathrm{d}z \\ &= \sum_{n \geq 1} \frac{z^n}{2n\pi i} \cdot 2\pi i \cdot \frac{n^{n-1}}{(n-1)!}, \\ a_n &= n^{n-1}.\end{aligned}$$

For part (c), since $a_n = n \cdot t_n$, we have

$$t_n = n^{n-2}.$$

Problem 4.13

You are designing an AI for a game based on tree search. For each state of the game, there are m possible actions, each of which leads to a new state of the game. At each state, the algorithm either stops and evaluates the current state, or expands it into exactly m child states and then recursively explores each of them.

In practice, the stopping decision may depend on various factors, such as the depth of the search, a time limit, or the quality of the estimate. In this problem, however, we assume a simple randomized model: at each state, the algorithm continues exploring with probability p, and stops with probability $1 - p$. All decisions are made independently across different states. Let T be the total number of states explored by the algorithm.

(a) Let $G(z)$ be the probability generating function of T. Show that

$$G(z) = z\left(1 - p + pG(z)^m\right).$$

(b) Determine a closed-form formula for $\Pr[T = t]$ for every $t \geq 0$.

Answer:

For part (a), the generating function contribution for stopping at the current state is $(1 - p)z$, while the contribution for continuing exploration is $pG(z)^m$, since independent subtrees correspond to convolution in the generating function. That is,

$$G(z) = z(1 - p + pG(z)^m).$$

For part (b), we define the holomorphic functions f and g such that $g(z) = G(z) = \sum_{t\geq 1} \Pr[T = t]z^t$, and $f(z)$ is the inverse function of $g(z)$, which means $f(g(z)) = g(f(z)) = z$.

Since $g(w) = w(1 - p + p \cdot g(w)^m)$ for all w, substituting $w = f(z)$ gives $z = f(z) \cdot (1 - p + p \cdot z^m)$, so $f(z) = \frac{z}{1-p+pz^m}$. By the Lagrange inversion theorem we derived in Problem 4.11, we have:

$$\begin{aligned} g(z) &= \sum_{t\geq 1} \frac{z^t}{2t\pi i} \int_{C_r} \frac{(1 - p + pz^m)^t}{z^t} \mathrm{d}z \\ &= \sum_{n\geq 0} \frac{z^{nm+1}}{nm+1} \binom{nm+1}{n} (1-p)^{nm+1-n} p^n, \\ \Pr[T = t] &= \mathbb{1}_{[t\equiv 1 \pmod m]} \cdot \frac{1}{t} \binom{t}{(t-1)/m} (1-p)^{t-(t-1)/m} p^{(t-1)/m}. \end{aligned}$$

Problem 4.14

Use Cauchy's Residue Theorem (see Remark 4.75) to evaluate the following integrals:

(a) $\displaystyle\int_{-\infty}^{\infty} \frac{\cos x}{x^2+4}\mathrm{d}x$;

(b) $\displaystyle\int_{-\infty}^{\infty} \frac{2x^2-1}{x^4+5x^2+4}\mathrm{d}x$;

(c) $\displaystyle\int_{-\infty}^{\infty} \frac{e^{x/3}}{e^x+1}\mathrm{d}x$.

Answer:

For part (a), let $f(z)=\frac{e^{iz}}{z^2+4}$, where $\Re(f(x))$ is exactly the integrand $\frac{\cos x}{x^2+4}$ on the real axis.

For each $R>2$, let Γ_R be the upper semicircle of radius R centered at the origin, and let γ_R be the closed contour formed by concatenating the interval $[-R,R]$ with Γ_R, oriented counterclockwise. Then we have

$$\int_{-R}^{R} f(x)\mathrm{d}x+\int_{\Gamma_R} f(z)\mathrm{d}z = 2\pi i\sum_{z_0}\mathrm{Res}(f,z_0),$$

where z_0 runs over all isolated singularities of f inside γ_R.

For each $z\in\Gamma_R$, we have

$$|f(z)|\le\frac{|e^{iz}|}{|z^2+4|}\le\frac{1}{R^2-4}.$$

Since the length of Γ_R is $L(\Gamma_R)=\pi R$, the ML inequality gives

$$\left|\int_{\Gamma_R} f(z)\mathrm{d}z\right|\le\frac{\pi R}{R^2-4}\to 0 \quad \text{as } R\to\infty.$$

The only singularity inside γ_R is the simple pole at $z=2i$, and

$$\mathrm{Res}(f,2i)=\frac{e^{i\cdot 2i}}{2\cdot 2i}=\frac{1}{4e^2 i}.$$

Taking the real part of both sides as $R\to\infty$, we obtain

$$\int_{-\infty}^{\infty}\frac{\cos x}{x^2+4}\mathrm{d}x=\Re\left(2\pi i\,\mathrm{Res}(f,2i)\right)=\frac{\pi}{2e^2}.$$

For part (b), let $f(z)=\frac{2z^2-1}{z^4+5z^2+4}$. We apply a similar argument as in part (a). Since the degree of the denominator is 2 greater than the degree of the numerator, $\lim_{R\to\infty}\int_{\Gamma_R} f(z)\mathrm{d}z=0$ by the ML inequality. The singularities inside γ_R are the simple poles at i and $2i$, and

$$\mathrm{Res}(f,i)=\frac{2i^2-1}{4i^3+10i}=-\frac{1}{2i},\quad \mathrm{Res}(f,2i)=\frac{2\cdot(2i)^2-1}{4\cdot(2i)^3+10\cdot(2i)}=\frac{3}{4i}.$$

Problem 4.14 continued on next page...

Therefore,

$$\int_{-\infty}^{\infty} \frac{2x^2-1}{x^4+5x^2+4}\mathrm{d}x = 2\pi i\left[\mathrm{Res}(f,i)+\mathrm{Res}(f,2i)\right] = \frac{\pi}{2}.$$

For part (c), let $f(z)=\frac{e^{z/3}}{e^z+1}$. Let $\gamma_R=\gamma_R^{(a)}+\gamma_R^{(b)}+\gamma_R^{(c)}+\gamma_R^{(d)}$ be a rectangular contour enclosing the rectangle $[-R,R]\times[0,2\pi]$, where $\gamma_R^{(a)}$ is the segment along the real axis from $-R$ to R, and $\gamma_R^{(b)},\gamma_R^{(c)},\gamma_R^{(d)}$ are the remaining three sides, oriented counterclockwise.

For each $z\in\gamma_R^{(b)}$ (the right vertical side), we have

$$|f(z)| \le \frac{e^{R/3}}{e^R-1}.$$

For each $z\in\gamma_R^{(d)}$ (the left vertical side), we have

$$|f(z)| \le \frac{e^{-R/3}}{1-e^{-R}}.$$

Since $L\left(\gamma_R^{(b)}\right)=L\left(\gamma_R^{(d)}\right)=2\pi$, the ML inequality gives

$$\left|\int_{\gamma_R^{(b)}} f(z)\mathrm{d}z\right| \le 2\pi\cdot\frac{e^{R/3}}{e^R-1}\to 0 \quad\text{and}\quad \left|\int_{\gamma_R^{(d)}} f(z)\mathrm{d}z\right| \le 2\pi\cdot\frac{e^{-R/3}}{1-e^{-R}}\to 0 \quad \text{as } R\to\infty.$$

Using the periodicity $e^{z+2\pi i}=e^z$, we have

$$f(x+2\pi i)=\frac{e^{(x+2\pi i)/3}}{e^{x+2\pi i}+1}=e^{2\pi i/3}f(x).$$

Since $\gamma_R^{(c)}$ is traversed from R to $-R$, this implies

$$\int_{\gamma_R^{(c)}} f(z)\mathrm{d}z = -e^{2\pi i/3}\int_{\gamma_R^{(a)}} f(z)\mathrm{d}z.$$

The only singularity inside γ_R is the simple pole at $z=\pi i$, and $\mathrm{Res}(f,\pi i)=-e^{\pi i/3}$. Therefore, as $R\to\infty$,

$$\int_{\gamma_R^{(a)}} f(z)\mathrm{d}z+\int_{\gamma_R^{(c)}} f(z)\mathrm{d}z = 2\pi i\,\mathrm{Res}(f,\pi i),$$

which gives

$$\int_{-\infty}^{\infty}\frac{e^{x/3}}{e^x+1}\mathrm{d}x=\frac{-2\pi i e^{\pi i/3}}{1-e^{2\pi i/3}}=\frac{2\sqrt{3}}{3}\pi.$$

Problem 4.15

For every odd integer $n \geq 1$, a permutation $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n) \in \mathrm{Perm}(n)$ is called an up-down permutation if

$$\sigma_1 < \sigma_2 > \sigma_3 < \sigma_4 > \cdots < \sigma_{n-1} > \sigma_n.$$

Let a_n be the number of up-down permutations of $[n]$. Then $a_1 = 1$ and $a_3 = 2$.

(a) Determine a closed-form formula for the exponential generating function of a_n

$$A(x) = \sum_{n \text{ odd}} a_n \frac{x^n}{n!}.$$

(b) Use Cauchy's Residue Theorem (see Remark 4.80) to express a_n as a simple function of the Riemann zeta function.

Answer:

For part (a), we consider the position of the largest element n. Since n is greater than all other elements, it must be placed at a peak position, which are exactly the even indices in an up-down permutation of odd length. Fixing n at any peak position splits the permutation into two independent up-down permutations on the left and right. This combinatorial decomposition translates to the differential equation:

$$A'(x) = 1 + A(x)^2.$$

Solving this separable differential equation:

$$\begin{aligned} \int \frac{A'(x)}{1 + A(x)^2} \mathrm{d}x &= \int \mathrm{d}x, \\ \int \frac{1}{1 + A(x)^2} \mathrm{d}A(x) &= \int \mathrm{d}x, \\ \arctan A(x) &= x + C. \end{aligned}$$

Since $A(0) = 0$, we have $C = 0$, so

$$A(x) = \tan x.$$

For part (b), the function $f(z) = \tan z$ is holomorphic at 0, and its singularities are the simple poles at $z = \left(k + \frac{1}{2}\right)\pi$ for all $k \in \mathbb{Z}$. Let $g_n(z) = \frac{f(z)}{z^{n+1}}$, then we have $a_n = n! \cdot \mathrm{Res}(g_n, 0)$ by the Cauchy integral formula for Taylor coefficients. Let $R_N = \left(N + \frac{3}{4}\right)\pi$, and let γ_N be the circular contour $|z| = R_N$ oriented counterclockwise. We first show that

$$\lim_{N \to \infty} \int_{\gamma_N} g_n(z) \mathrm{d}z = 0.$$

For any complex number $z = x + iy$, we have

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}, \quad \cos z = \frac{e^{iz} + e^{-iz}}{2},$$

and the standard modulus identities:

$$|\sin z|^2 = \sin^2 x + \sinh^2 y, \quad |\cos z|^2 = \cos^2 x + \sinh^2 y.$$

Problem 4.15 continued on next page...

Therefore,

$$|\tan z|^2 = \frac{\sin^2 x + \sinh^2 y}{\cos^2 x + \sinh^2 y} = 1 - \frac{\cos 2x}{\cos^2 x + \sinh^2 y} \le 1 + \frac{1}{\sinh^2 y} = \coth^2 |y|.$$

For each $z \in \gamma_N$, write $z = R_N e^{i\theta}$, so $x = R_N \cos\theta$ and $y = R_N \sin\theta$. We consider two cases:

- If $|y| \ge 1$, then $|\tan z| \le \coth 1$, which is a constant independent of N.
- If $|y| < 1$, then $|\cos\theta| = \frac{\sqrt{R_N^2 - y^2}}{R_N} \ge 1 - o(1)$ as $N \to \infty$. In this case,

$$|\sin z| = \sqrt{\sin^2 x + \sinh^2 y} \le \sqrt{1 + \sinh^2 1},$$

$$|\cos z| \ge |\cos(R_N \cos\theta)| \ge |\cos R_N| - o(1) = \frac{\sqrt{2}}{2} - o(1),$$

  which implies $|\tan z| = \frac{|\sin z|}{|\cos z|} \le \frac{\sqrt{1+\sinh^2 1}}{\frac{\sqrt{2}}{2} - o(1)}$, bounded by a constant for sufficiently large N.

In conclusion, there exists a constant $M > 0$ independent of N such that $|\tan z| \le M$ for all $z \in \gamma_N$, so

$$|g_n(z)| = \frac{|\tan z|}{|z|^{n+1}} \le \frac{M}{R_N^{n+1}} = \mathcal{O}(N^{-n-1}).$$

Since the length of γ_N is $L(\gamma_N) = 2\pi R_N = \mathcal{O}(N)$, the ML inequality gives

$$\left| \int_{\gamma_N} g_n(z) \mathrm{d}z \right| \le \frac{2\pi M}{R_N^n} \to 0 \quad \text{as } N \to \infty.$$

By Cauchy's Residue Theorem, this gives

$$a_n = n! \cdot \mathrm{Res}(g_n, 0) = -n! \cdot \sum_{k=-\infty}^{\infty} \mathrm{Res}(g_n, (k+1/2)\pi).$$

Each pole $z = \left(k + \frac{1}{2}\right)\pi$ is a simple pole of g_n. Using the simple pole residue formula, we have

$$\mathrm{Res}(g_n, (k+1/2)\pi) = \lim_{z \to (k+1/2)\pi} \frac{(z - (k+1/2)\pi)\sin z}{z^{n+1}\cos z} = -\frac{1}{((k+1/2)\pi)^{n+1}}.$$

Substituting back, we get

$$\begin{aligned} a_n &= \frac{n!}{\pi^{n+1}} \cdot \sum_{k=-\infty}^{\infty} \frac{1}{(k+1/2)^{n+1}} \\ &= \frac{n!}{\pi^{n+1}} \cdot 2 \sum_{k \ge 0} \frac{1}{(k+1/2)^{n+1}} \\ &= \frac{2^{n+2} n!}{\pi^{n+1}} \cdot \sum_{k \ge 0} \frac{1}{(2k+1)^{n+1}} \\ &= \frac{2n!}{\pi^{n+1}} \cdot \left(2^{n+1} - 1\right) \zeta(n+1). \end{aligned}$$

Problem 4.16

Write a complete proof of Lemma 4.87: Let $c > 0$. For every $\delta \in \mathbb{R}$, we have

$$\lim_{T\to+\infty} \frac{1}{2\pi i} \int_{\gamma_{c,T}(t)} \frac{e^{\delta z}}{z} \mathrm{d}z = \begin{cases} 0 & \text{if } \delta < 0, \\ 1/2 & \text{if } \delta = 0, \\ 1 & \text{if } \delta > 0, \end{cases}$$

where $\gamma_{c,T}(t) = c + it$ for $t \in [-T, T]$.

Answer:

Let $f(z) = \frac{e^{\delta z}}{z}$. Then $f(z)$ is holomorphic on $\mathbb{C} \setminus \{0\}$, and 0 is a simple pole with residue 1. Define $I_x(T) = \int_{\gamma_{x,T}} f(z)\mathrm{d}z$ for any real number x. Our aim is to compute $\lim_{T\to+\infty} I_c(T)$.

- If $\delta = 0$, we directly integrate $f(z)$ over $\gamma_{c,T}$.

  $$\int_{\gamma_{c,T}} f(z)\mathrm{d}z = \int_{-T}^{T} f(\gamma_{c,T}(t))\gamma'_{c,T}(t)\mathrm{d}t = \int_{-T}^{T} \frac{i}{c+it}\mathrm{d}t = \int_{-T}^{T} \frac{ic+t}{c^2+t^2}\mathrm{d}t = 2i \arctan\left(\frac{T}{c}\right),$$

  while $\arctan\left(\frac{T}{c}\right) \to \frac{\pi}{2}$ as $T \to \infty$, which implies $I_c(T) \to \pi i$.

- If $\delta < 0$, let a be a real number such that $a > c$. Let $\gamma_{c,a,T} = \gamma_{a,T} + \gamma^{(b)} + \gamma'_{c,T} + \gamma^{(d)}$ be a rectangular contour enclosing the rectangle $[c, a] \times [-T, T]$, oriented counterclockwise, where $\gamma'_{c,T}$ is the reverse of $\gamma_{c,T}$.

  Since the contour encloses no singularities of $f(z)$, by Cauchy's theorem, the integral around $\gamma_{c,a,T}$ is zero. Rearranging gives:

  $$I_c(T) = I_a(T) - \int_{\gamma^{(b)}} f(z)\mathrm{d}z - \int_{\gamma^{(d)}} f(z)\mathrm{d}z.$$

  For each $z \in \gamma^{(b)}$, write $z = x + iT$. Then

  $$|f(z)| = \frac{|e^{\delta z}|}{|z|} = \frac{e^{\delta x}}{\sqrt{x^2+T^2}} \le \frac{e^{\delta x}}{T}.$$

  Therefore,

  $$\left|\int_{\gamma^{(b)}} f(z)\mathrm{d}z\right| \le \int_c^a |f(z)||\mathrm{d}z| \le \int_c^a \frac{e^{\delta x}}{T}\mathrm{d}x = \frac{e^{\delta a} - e^{\delta c}}{\delta T}.$$

  Since $\delta < 0$ and $a > c$, we have $|e^{\delta a} - e^{\delta c}| \le e^{\delta c}$, a constant independent of T and a. Hence this integral tends to 0 as $T \to \infty$.

  The same argument applies to $\gamma^{(d)}$.

  For the vertical integral $I_a(T)$, write $z = a + iy$ for $y \in [-T, T]$. Then:

  $$|I_a(T)| = \left|\int_{-T}^{T} \frac{e^{\delta(a+iy)}}{a+iy} i\mathrm{d}y\right| \le e^{\delta a} \int_{-T}^{T} \frac{1}{\sqrt{a^2+y^2}}\mathrm{d}y = 2e^{\delta a} \ln\left(\frac{T}{a} + \sqrt{1+\left(\frac{T}{a}\right)^2}\right).$$

  For any fixed T, as $a \to +\infty$, the logarithmic term tends to $\ln(1) = 0$, and $e^{\delta a} \to 0$ exponentially fast. Thus $\lim_{a\to+\infty} I_a(T) = 0$ for all T.

Problem 4.16 continued on next page...

Taking the limit $T \to \infty$ followed by $a \to +\infty$, we conclude:

$$\lim_{T\to\infty} I_c(T) = 0.$$

- If $\delta > 0$, let a be a real number such that $a < 0$. Let $\gamma_{a,c,T} = \gamma'_{a,T} + \gamma^{(b)} + \gamma_{c,T} + \gamma^{(d)}$ be a rectangular contour enclosing the rectangle $[a, c] \times [-T, T]$, oriented counterclockwise, where $\gamma'_{a,T}$ is the reverse of $\gamma_{a,T}$.

  This contour encloses the simple pole of $f(z)$ at 0. By the residue theorem, the integral around $\gamma_{a,c,T}$ is $2\pi i \cdot \mathrm{Res}(f, 0) = 2\pi i$. Rearranging gives:

$$I_c(T) = 2\pi i + I_a(T) - \int_{\gamma^{(b)}} f(z)\mathrm{d}z - \int_{\gamma^{(d)}} f(z)\mathrm{d}z.$$

  Let $a \to -\infty$. The same arguments as in the $\delta < 0$ case show that the integrals along $\gamma^{(b)}$, $\gamma^{(d)}$, and $\gamma_{a,T}$ are all negligible. Taking the limit $T \to \infty$ followed by $a \to -\infty$, we obtain:

$$\lim_{T\to\infty} I_c(T) = 2\pi i.$$

Dividing each of these results by $2\pi i$ completes the proof of the lemma.

Problem 4.17

In this problem, we will guide you through the proof of Cramér's theorem (Theorem 3.36). For any probability mass function p, we write $p^{(\lambda)}$ for the probability mass function of the λ-tilted distribution of p. That is,

$$p^{(\lambda)}(x) = \frac{p(x)e^{\lambda x}}{\sum_y p(y)e^{\lambda y}}, \forall x \in \mathbb{R}.$$

Let D be a probability distribution with probability mass function p, minimum value ℓ, maximum value u, mean μ, MGF $M(t)$ and rate function $I(a)$. Let $X_1, \ldots, X_n$ be i.i.d. random variables drawn from D, and let $X^{(n)} := \frac{1}{n}\sum_{i=1}^n X_i$ be their sample average.

For simplicity, we restrict attention to the case $a \in (\mu, u)$. Fix such an a, and our goal is to show that as $n \to \infty$,

$$\Pr\left[X^{(n)} \geq a\right] = e^{-n(I(a)+o(1))}.$$

(a) Let $\lambda \in \mathbb{R}$ be a parameter. Let $Y_1, \ldots, Y_n$ be i.i.d. random variables, each with distribution $q = p^{(\lambda)}$. Let p_n be the probability mass function of $X^{(n)} := \frac{1}{n}\sum_{i=1}^n X_i$ and q_n be the probability mass function of $Y^{(n)} := \frac{1}{n}\sum_{i=1}^n Y_i$. Show that for all $x \in \mathbb{R}$

$$q_n(x) = p_n^{(n\lambda)}(x).$$

(b) Prove that for any $\lambda \in \mathbb{R}$, the corresponding $Y^{(n)}$ satisfies

$$\Pr\left[X^{(n)} \geq a\right] = M(\lambda)^n \cdot \mathbb{E}\left[\mathbb{1}_{\left[Y^{(n)} \geq a\right]} e^{-n\lambda Y^{(n)}}\right].$$

(c) For every $\epsilon > 0$ such that $a + \epsilon < u$, show that it is possible to find a value $\lambda = \lambda(\epsilon) > 0$ such that

Problem 4.17 continued on next page...

$\mathbb{E}[Y^{(n)}] = a + \epsilon$. Then show that

$$\lim_{n\to\infty} \Pr\left[a \le Y^{(n)} \le a + \epsilon + \epsilon/\lambda\right] = 1.$$

(d) Prove Cramér's theorem for $a \in (\mu, u)$. Since Chernoff bounds already give us the upper bound $\Pr[X^{(n)} \ge a] \le e^{-nI(a)}$, you only need to show the following lower bound:

$$\log \Pr\left[X^{(n)} \ge a\right] \ge -n(I(a) + o(1)),$$

or equivalently,

$$\liminf_{n\to\infty} \frac{1}{n} \log \Pr\left[X^{(n)} \ge a\right] \ge -I(a).$$

Answer:

For part (a), we denote the moment generating functions of $X_i, Y_i, X^{(n)}, Y^{(n)}$ as $M_X(t), M_Y(t), M_{X^{(n)}}(t), M_{Y^{(n)}}(t)$, respectively. By property of exponential tilting, we have:

$$M_Y(t) = \frac{M_X(\lambda + t)}{M_X(\lambda)}.$$

By the property of MGFs, we have:

$$M_{X^{(n)}}(t) = M_X\left(\frac{t}{n}\right)^n,$$
$$M_{Y^{(n)}}(t) = M_Y\left(\frac{t}{n}\right)^n = \left(\frac{M_X\left(\lambda + t/n\right)}{M_X(\lambda)}\right)^n.$$

Now consider the MGF of the $n\lambda$-tilted distribution of p_n, denoted $M_{p_n^{(n\lambda)}}(t)$:

$$M_{p_n^{(n\lambda)}}(t) = \frac{M_{X^{(n)}}(n\lambda + t)}{M_{X^{(n)}}(n\lambda)} = \frac{M_X\left(\lambda + t/n\right)^n}{M_X(\lambda)^n}.$$

We see that $M_{Y^{(n)}}(t) = M_{p_n^{(n\lambda)}}(t)$ for all t, which implies $q_n(x) = p_n^{(n\lambda)}(x)$ for all $x \in \mathbb{R}$.

For part (b), we start by rewriting the probability using the result from part (a). From the definition of exponential tilting, we have $p_n(x) = q_n(x) M_{X^{(n)}}(n\lambda) e^{-n\lambda x}$. Since $M_{X^{(n)}}(n\lambda) = M_X(\lambda)^n$, we obtain:

$$\begin{aligned}
\Pr\left[X^{(n)} \ge a\right] = \sum_{x \ge a} p_n(x) &= \sum_{x \ge a} q_n(x) M_X(\lambda)^n e^{-n\lambda x} \\
&= M_X(\lambda)^n \sum_x \mathbb{1}_{[x \ge a]} e^{-n\lambda x} q_n(x) \\
&= M_X(\lambda)^n \cdot \mathbb{E}\left[\mathbb{1}_{\left[Y^{(n)} \ge a\right]} e^{-n\lambda Y^{(n)}}\right].
\end{aligned}$$

For part (c), we denote the cumulant generating function of X_i as $K_X(t) = \log M_X(t)$. By the properties of CGFs and exponential tilting, the mean of the tilted distribution is:

$$\mathbb{E}[Y^{(n)}] = \mathbb{E}[Y_i] = K_X'(\lambda).$$

Since the original distribution D is non-degenerate, we have $K_X''(\lambda) = \mathrm{Var}(Y_i) > 0$, which implies that $\mathbb{E}[Y_i]$

is a strictly increasing function of λ. We now analyze the boundary behavior:

- When $\lambda \to 0^+$, we have $\mathbb{E}[Y_i] = K'_X(\lambda) \to K'_X(0) = \mathbb{E}[X_i] = \mu < a + \epsilon$.
- When $\lambda \to +\infty$, the exponential term $e^{\lambda u}$ dominates all other terms $e^{\lambda x}$ for $x < u$, so the probability mass concentrates at u, and thus $\mathbb{E}[Y_i] \to u > a + \epsilon$.

By the intermediate value theorem, there exists a unique $\lambda = \lambda(\epsilon) > 0$ such that $\mathbb{E}[Y^{(n)}] = a + \epsilon$.

Therefore, by Chebyshev's inequality, we have:

$$\Pr\left[Y^{(n)} \le a\right] = \Pr\left[Y^{(n)} \le \mathbb{E}[Y^{(n)}] - \epsilon\right] \le \frac{\mathrm{Var}(Y_i)}{n\epsilon^2} \to 0,$$
$$\Pr\left[Y^{(n)} \ge a + \epsilon + \epsilon/\lambda\right] = \Pr\left[Y^{(n)} \ge \mathbb{E}[Y^{(n)}] + \epsilon/\lambda\right] \le \frac{\lambda^2 \mathrm{Var}(Y_i)}{n\epsilon^2} \to 0.$$

Combining these results, we obtain:

$$\lim_{n\to\infty} \Pr\left[a \le Y^{(n)} \le a + \epsilon + \epsilon/\lambda\right] = 1.$$

For part (d), fix any $\epsilon > 0$ such that $a + \epsilon < u$, and let $\lambda = \lambda(\epsilon) > 0$ be the value from part (c) satisfying $\mathbb{E}[Y^{(n)}] = a + \epsilon$. From part (b), we have:

$$\frac{1}{n} \log \Pr\left[X^{(n)} \ge a\right] = K_X(\lambda) + \frac{1}{n} \log \mathbb{E}\left[\mathbb{1}_{\left[Y^{(n)} \ge a\right]} e^{-n\lambda Y^{(n)}}\right].$$

Using the concentration result from part (c), we can bound the expectation from below:

$$\mathbb{E}\left[\mathbb{1}_{\left[Y^{(n)} \ge a\right]} e^{-n\lambda Y^{(n)}}\right] \ge \Pr\left[a \le Y^{(n)} \le a + \epsilon + \epsilon/\lambda\right] \cdot e^{-n\lambda(a+\epsilon+\epsilon/\lambda)}.$$

Taking logarithms and dividing by n, we get:

$$\frac{1}{n} \log \mathbb{E}\left[\mathbb{1}_{\left[Y^{(n)} \ge a\right]} e^{-n\lambda Y^{(n)}}\right] \ge \frac{1}{n} \log \Pr\left[a \le Y^{(n)} \le a + \epsilon + \epsilon/\lambda\right] - \lambda\left(a + \epsilon + \epsilon/\lambda\right).$$

As $n \to \infty$, the first term on the right-hand side tends to 0 (since the probability tends to 1). Therefore:

$$\liminf_{n\to\infty} \frac{1}{n} \log \Pr\left[X^{(n)} \ge a\right] \ge K_X(\lambda) - \lambda(a + \epsilon) - \epsilon.$$

By the definition of the rate function, $I(a+\epsilon) = \sup_{t\in\mathbb{R}}\{t(a+\epsilon) - K_X(t)\} \ge \lambda(a+\epsilon) - K_X(\lambda)$, which implies $K_X(\lambda) - \lambda(a + \epsilon) \ge -I(a + \epsilon)$. Substituting this in, we obtain:

$$\liminf_{n\to\infty} \frac{1}{n} \log \Pr\left[X^{(n)} \ge a\right] \ge -I(a + \epsilon) - \epsilon.$$

Since this inequality holds for all sufficiently small $\epsilon > 0$, and the rate function $I(a)$ is continuous at the interior point $a \in (\mu, u)$, we can take the limit as $\epsilon \to 0^+$ to derive the desired bound:

$$\liminf_{n\to\infty} \frac{1}{n} \log \Pr\left[X^{(n)} \ge a\right] \ge -I_X(a).$$

Combining this lower bound with the Chernoff upper bound completes the proof of Cramér's theorem.